Maaz Bin Fazal

GitHub | LinkedIn | www.maazbinfazal.engineer | engr.maazbinfazal.ee@gmail.com |
+92.34.007.31861

Awards and Honors

- Ranked Top 75 globally in UC Berkeley's Coding Challenge (1000+ teams; 2,231 contestants).
- Recognized for strong performance in **Meta worldwide** algorithmic programming contest.
- Delivered one journal and one conference **(IEEE Xplore)** publication advancing my research area.
- Received Merit Scholarship from the state in university.
- Scored 120/160 in Duolingo English Test (IELTS **6.5/9** equivalent).

Education

Islamia University of Bahawalpur July 2021 – June 2025
Bachelor of **Electronics Engineering** CGPA: 3.31/4.00 Bahawalpur, Pakistan

Relevant Coursework: Transcript Artificial Intelligence, Computer Architecture, Multivariable Calculus, Applied Physics, Robotics, Power Electronics, Signals & Systems, Very Large-Scale Integration (VLSI), Programming Fundamentals, Control Systems, System Design.

FYP (Research Work) Prototype
Automated Fruit Quality Detection and Sorting Using Real-Time Deep Learning Image Processing

- **Optimized YOLOv8 architecture** for enhanced real time fruit quality detection accuracy.
- Developed & deployed an optimized deep learning model achieving **90%+ mAP**.
- Designed automated conveyor sorting with servo control to **reduce manual labor in agriculture**.

Publications

[C1]. **Maaz Bin Fazal**, Muhammad Inam Ul Haq, S. Mehmood, S. Sheikh, Zeeshan, Muhammad Sohail, "Low-Cost Real Time Fruit Quality Assessment and Sorting Using YOLOv8," in *Proc. 8th Global Conference on Wireless & Optical Technologies (GCWOT)*, IEEE Xplore indexed, Malaga, Spain, Feb. 2026, pp. 1–6. url: PDF

[J1]. **Maaz Bin Fazal**, Muhammad Inam Ul Haq, Muhammad Sohail, "Design and Implementation of an AC–DC Converter for Efficient Power Conversion," *Spectrum of Engineering*, vol. 13, no. 2, pp. 1–12, Nov. 2025. **(Accepted for publication)** url: PDF

[J2]. Muhammad Inam Ul Haq, **Maaz Bin Fazal**, Z. Arfeen, "Q-INTEL: Hybrid Quantum Classical Intelligence for Autonomous Engineering Systems," *Computer of Engineering*, under first-round review, 2025. url: PDF

[J3]. Z. Arfeen, Muhammad Inam Ul Haq, **Maaz Bin Fazal**, "Trust Enhanced RISC-V IEEE-754 FPU: ECC, Redundancy, and Firewall Protected Execution for SweRV EH1," Manuscript in progress, 2025. url: PDF

Work Experience

Assistant to Hardware Engineer, CARE Pakistan Jun. 2022 – May. 2024

- Assisted in tracing circuits for **Pakistan Army helicopter** battery chargers and autoclave systems.
- Supported in programming microcontrollers for **electric bikes** and various embedded mini projects.

Research Assistant,Deep Embedded Lab Pakistan Jun. 2024 – Jun. 2025

- Engaged in technical writing for a research paper, supporting its successful publication.
- Optimized YOLO through hyperparameter tuning, achieving 90% mAP accuracy.

Teaching Assistant, Computer Architecture

- Helped students understand tough concepts like pipelining, memory hierarchy, and cache design.
- Improved my teaching and communication skills by explaining ideas in clear, simple ways.

Volunteer Experience

- Provided virtual guidance to school and college students on STEM pathways.
- Taught foundational software skills and basic Python programming.
- Led 50+ volunteers statewide to prepare 500+ underprivileged students for university entrance exams.

International Hackathons

Robo Styler AI Agent **Demo | GitHub**

- Built an **AI Roblox styling agent** using Coral, Mistral, Firecrawl, and ElevenLabs.
- Built a Node.js backend integrated with Roblox Studio and registered the agent in Coral Registry.
- Deployed on Nebius AI, enabling **real-time outfit generation and marketplace transactions**.

AI Content Wallet – Arc Blockchain Micro-Payments **Demo | GitHub**

- Built an **AI agent** that analyzes user activity and automates USDC micro-payments on Arc's blockchain.
- Developed an off-chain backend (TypeScript/Node.js / Cloudflare Workers) enabling **sub-second payments** and auto-subscriptions.
- Enabled global creator payouts with realtime USD transfers for users/creators across emerging markets.

Academic Projects

Students Performance Prediction System Demo | Report | Code

- Built a **predictive system** using ML algorithms (Random Forest, SVM, Logistic Regression).
- Deployed the model on Render with GitHub integration for **web access**.
- Enabled **early detection of at-risk students** for academic support.

Home Automation System Report

- Designed **IoT-based smart home automation** using Arduino and mobile app.
- Implemented **real-time monitoring** and remote control via Wi-Fi.
- Delivered an **energy-efficient and user-friendly** automation solution.

Industrial Automation with PLC Tasks | Demo

- Programmed Siemens and Fatek PLCs for automation of motors and processes.
- Developed ladder logic for water tank level monitoring and traffic signals.
- Created **small-scale demos simulating real-world industrial automation**.

Vending Machine (Verilog) Report

- Designed a vending machine in **Verilog HDL** and implemented on **Spartan XC400 FPGA**.
- Added coin recognition and purchase logic for digital transactions.
- Improved convenience and fairness through **optimized hardware design**.

Skills

Programming Languages	**Python**, C++, Assembly Language, VHDL
Web Development	HTML5, JavaScript, React.js, Tailwind CSS, Node.js, REST APIs
Libraries & Frameworks	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Overleaf/LaTeX
Simulation & Design Tools	MATLAB/Simulink, Proteus, Viper, Power Esim, Tinkercad, CubeMX
Software & IDEs	Keil, Arduino IDE, VS Code, PyCharm, Google Colab, Jupyter Notebook, Microsoft 365, Meero
Version Control	Git, GitHub